

The data below are from a sample of patients with insomnia problems in a double-blind clinical trial comparing an active hypnotic drug with a placebo. The response is the patient's reported time (in minutes) to fall asleep after going to bed. Patients responded before and following a 2-week treatment period. The two treatments, active drug and placebo, form a binary explanatory variable. The subjects were randomly allocated to the treatment groups. Here, each subject forms a cluster, with the observations in a cluster being the ordinal response at the two occasions of observation.

The response is 1, 2, 3, or 4, which denote the time to sleep was <20, 20–30, 30–60, and >60 minutes respectively.

```
data sleep;
input case treat occasion outcome;
datalines;
1      1      0      1
1      1      1      1
2      1      0      1
2      1      1      1
3      1      0      1
.....
236    0      0      4
236    0      1      4
237    0      0      4
237    0      1      4
238    0      0      4
238    0      1      4
239    0      0      4
239    0      1      4
;
```

We are going to fit this data with

$$\log \left\{ \frac{P(Y_{ij} \leq k | b_{i0})}{P(Y_{ij} > k | b_{i0})} \right\} = \beta_0 + \beta_1 G_i + \beta_2 t_{ij} + \beta_3 G_i t_{ij} + b_{i0}$$

The effect of going from baseline to post treatment is  $e^{\beta_2}$  for the control group and  $e^{\beta_2 + \beta_3}$  for the treatment group.

```
proc glimmix method=quad(qpoints=50) data=sleep;
class case;
model outcome = treat occasion treat*occasion / link=cumlogit
dist=multinomial solution;
random int / subject=case;
run;
```

|                |
|----------------|
| The SAS System |
|----------------|

The GLIMMIX Procedure

**Model Information**

**Data Set**

WORK.SLEEP

### Model Information

|                                   |                          |
|-----------------------------------|--------------------------|
| <b>Response Variable</b>          | outcome                  |
| <b>Response Distribution</b>      | Multinomial (ordered)    |
| <b>Link Function</b>              | Cumulative Logit         |
| <b>Variance Function</b>          | Default                  |
| <b>Variance Matrix Blocked By</b> | case                     |
| <b>Estimation Technique</b>       | Maximum Likelihood       |
| <b>Likelihood Approximation</b>   | Gauss-Hermite Quadrature |
| <b>Degrees of Freedom Method</b>  | Containment              |

### Class Level Information

#### Class Levels Values

case 239 1 2 3 4 5 6 7 8 9... 239

**Number of Observations Read** 478

**Number of Observations Used** 478

### Response Profile

| <b>Ordered Value</b> | <b>outcome</b> | <b>Total Frequency</b> |
|----------------------|----------------|------------------------|
| 1                    | 1              | 97                     |
| 2                    | 2              | 118                    |
| 3                    | 3              | 129                    |
| 4                    | 4              | 134                    |

The GLIMMIX procedure is modeling the probabilities of levels of outcome having lower Ordered Values in the Response Profile table.

### Dimensions

|                                 |     |
|---------------------------------|-----|
| <b>G-side Cov. Parameters</b>   | 1   |
| <b>Columns in X</b>             | 6   |
| <b>Columns in Z per Subject</b> | 1   |
| <b>Subjects (Blocks in V)</b>   | 239 |

**Dimensions****Max Obs per Subject** 2**Optimization Information**

|                                   |                   |
|-----------------------------------|-------------------|
| <b>Optimization Technique</b>     | Dual Quasi-Newton |
| <b>Parameters in Optimization</b> | 7                 |
| <b>Lower Boundaries</b>           | 1                 |
| <b>Upper Boundaries</b>           | 0                 |
| <b>Fixed Effects</b>              | Not Profiled      |
| <b>Starting From</b>              | GLM estimates     |
| <b>Quadrature Points</b>          | 50                |

**Iteration History**

| <b>Iteration</b> | <b>Restarts</b> | <b>Evaluations</b> | <b>Objective<br/>Function</b> | <b>Change</b> | <b>Max<br/>Gradient</b> |
|------------------|-----------------|--------------------|-------------------------------|---------------|-------------------------|
| 0                | 0               | 4                  | 1226.9387087                  | .             | 72.44447                |
| 1                | 0               | 2                  | 1196.3631316                  | 30.57557711   | 18.10421                |
| 2                | 0               | 3                  | 1193.0623213                  | 3.30081038    | 20.91976                |
| 3                | 0               | 2                  | 1190.3894351                  | 2.67288616    | 13.57284                |
| 4                | 0               | 3                  | 1189.2483069                  | 1.14112818    | 6.703743                |
| 5                | 0               | 2                  | 1187.5023462                  | 1.74596070    | 2.778815                |
| 6                | 0               | 2                  | 1186.6273329                  | 0.87501327    | 4.7954                  |
| 7                | 0               | 2                  | 1186.2377183                  | 0.38961463    | 3.348319                |
| 8                | 0               | 2                  | 1186.071747                   | 0.16597136    | 2.1565                  |
| 9                | 0               | 2                  | 1185.9977503                  | 0.07399664    | 1.405478                |
| 10               | 0               | 2                  | 1185.9726747                  | 0.02507558    | 1.036631                |
| 11               | 0               | 2                  | 1185.94461                    | 0.02806477    | 0.104798                |
| 12               | 0               | 3                  | 1185.9442356                  | 0.00037435    | 0.012845                |
| 13               | 0               | 3                  | 1185.9442054                  | 0.00003020    | 0.002786                |
| 14               | 0               | 3                  | 1185.944205                   | 0.00000045    | 0.000949                |

Convergence criterion (GCONV=1E-8) satisfied.

### Fit Statistics

|                                 |         |
|---------------------------------|---------|
| <b>-2 Log Likelihood</b>        | 1185.94 |
| <b>AIC (smaller is better)</b>  | 1199.94 |
| <b>AICC (smaller is better)</b> | 1200.18 |
| <b>BIC (smaller is better)</b>  | 1224.28 |
| <b>CAIC (smaller is better)</b> | 1231.28 |
| <b>HQIC (smaller is better)</b> | 1209.75 |

### Fit Statistics for Conditional Distribution

|                                       |        |
|---------------------------------------|--------|
| <b>-2 log L(outcome   r. effects)</b> | 789.00 |
|---------------------------------------|--------|

### Covariance Parameter Estimates

| <b>Cov Parm</b>  | <b>Subject</b> | <b>Estimate</b> | <b>Standard Error</b> |
|------------------|----------------|-----------------|-----------------------|
| <b>Intercept</b> | <b>case</b>    | 3.6280          | 0.8815                |

### Solutions for Fixed Effects

| <b>Effect</b>         | <b>outcome</b> | <b>Estimate</b> | <b>Standard Error</b> | <b>DF</b> | <b>t Value</b> | <b>Pr &gt;  t </b> |
|-----------------------|----------------|-----------------|-----------------------|-----------|----------------|--------------------|
| <b>Intercept</b>      | <b>1</b>       | -3.4896         | 0.3588                | 237       | -9.73          | <.0001             |
| <b>Intercept</b>      | <b>2</b>       | -1.4846         | 0.2903                | 237       | -5.11          | <.0001             |
| <b>Intercept</b>      | <b>3</b>       | 0.5613          | 0.2702                | 237       | 2.08           | 0.0388             |
| <b>treat</b>          |                | 0.05786         | 0.3663                | 235       | 0.16           | 0.8746             |
| <b>occasion</b>       |                | 1.6016          | 0.2834                | 235       | 5.65           | <.0001             |
| <b>treat*occasion</b> |                | 1.0813          | 0.3805                | 235       | 2.84           | 0.0049             |

### Type III Tests of Fixed Effects

| <b>Effect</b>         | <b>Num DF</b> | <b>Den DF</b> | <b>F Value</b> | <b>Pr &gt; F</b> |
|-----------------------|---------------|---------------|----------------|------------------|
| <b>treat</b>          | 1             | 235           | 0.02           | 0.8746           |
| <b>occasion</b>       | 1             | 235           | 31.95          | <.0001           |
| <b>treat*occasion</b> | 1             | 235           | 8.08           | 0.0049           |

Next we are going to fit the data with the baseline category logistic model:

$$\log \left\{ \frac{P(Y_{ij} = k | b_{i0})}{P(Y_{ij} = 1 | b_{i0})} \right\} = \beta_{0k} + \beta_{1k} G_i + \beta_{2k} t_{ij} + \beta_{3k} G_i t_{ij} + b_{i0k}$$

```
proc glimmix method=quad(qpoints=10) data=sleep;
class case outcome(ref="1");
model outcome = treat occasion treat*occasion / link=GLOGIT dist=multinomial
solution;
random int / subject=case group=outcome;
run;
```

The SAS System

The GLIMMIX Procedure

Model Information

|                            |                          |
|----------------------------|--------------------------|
| Data Set                   | WORK.SLEEP               |
| Response Variable          | outcome                  |
| Response Distribution      | Multinomial (nominal)    |
| Link Function              | Generalized Logit        |
| Variance Function          | Default                  |
| Variance Matrix Blocked By | case                     |
| Estimation Technique       | Maximum Likelihood       |
| Likelihood Approximation   | Gauss-Hermite Quadrature |
| Degrees of Freedom Method  | Containment              |

Class Level Information

| Class   | Levels | Values              |
|---------|--------|---------------------|
| case    | 239    | 1 2 3 4 5 6 ... 239 |
| outcome | 4      | 1 2 3 4             |

|                             |     |
|-----------------------------|-----|
| Number of Observations Read | 478 |
| Number of Observations Used | 478 |

Response Profile

| Ordered outcome Value | Total Frequency |
|-----------------------|-----------------|
| 1 1                   | 97              |

### Response Profile

| Ordered outcome<br>Value | Total<br>Frequency |
|--------------------------|--------------------|
| 2 2                      | 118                |
| 3 3                      | 129                |
| 4 4                      | 134                |

In modeling category probabilities, outcome='1' serves as the reference category.

### Dimensions

|                          |     |
|--------------------------|-----|
| G-side Cov. Parameters   | 3   |
| Columns in X             | 12  |
| Columns in Z per Subject | 3   |
| Subjects (Blocks in V)   | 239 |
| Max Obs per Subject      | 2   |

### Optimization Information

|                            |                   |
|----------------------------|-------------------|
| Optimization Technique     | Dual Quasi-Newton |
| Parameters in Optimization | 15                |
| Lower Boundaries           | 3                 |
| Upper Boundaries           | 0                 |
| Fixed Effects              | Not Profiled      |
| Starting From              | GLM estimates     |
| Quadrature Points          | 10                |

### Iteration History

| Iteration | Restarts | Evaluations | Objective<br>Function | Change      | Max<br>Gradient |
|-----------|----------|-------------|-----------------------|-------------|-----------------|
| 0         | 0        | 4           | 1239.7426277          | .           | 22.20949        |
| 1         | 0        | 2           | 1228.1019538          | 11.64067383 | 16.4115         |
| 2         | 0        | 3           | 1224.4588175          | 3.64313629  | 9.375126        |
| 23        | 0        | 3           | 1199.5074377          | 0.00001444  | 0.002029        |
| 24        | 0        | 3           | 1199.507437           | 0.00000075  | 0.001723        |

Convergence criterion (GCONV=1E-8) satisfied.

**Estimated G matrix is not positive definite.**

#### Fit Statistics

|                                 |         |
|---------------------------------|---------|
| <b>-2 Log Likelihood</b>        | 1199.51 |
| <b>AIC (smaller is better)</b>  | 1227.51 |
| <b>AICC (smaller is better)</b> | 1228.41 |
| <b>BIC (smaller is better)</b>  | 1276.18 |
| <b>CAIC (smaller is better)</b> | 1290.18 |
| <b>HQIC (smaller is better)</b> | 1247.12 |

#### Fit Statistics for Conditional Distribution

|                                       |        |
|---------------------------------------|--------|
| <b>-2 log L(outcome   r. effects)</b> | 826.62 |
|---------------------------------------|--------|

#### Covariance Parameter Estimates

| <b>Cov Parm</b>  | <b>Subject</b> | <b>Group</b>     | <b>Estimate</b> | <b>Standard Error</b> |
|------------------|----------------|------------------|-----------------|-----------------------|
| <b>Intercept</b> | <b>case</b>    | <b>outcome 2</b> | 0               | .                     |
| <b>Intercept</b> | <b>case</b>    | <b>outcome 3</b> | 0.7530          | 0.5949                |
| <b>Intercept</b> | <b>case</b>    | <b>outcome 4</b> | 6.4050          | 2.6088                |

#### Solutions for Fixed Effects

| <b>Effect</b>    | <b>outcome</b> | <b>Estimate</b> | <b>Standard Error</b> | <b>DF</b> | <b>t Value</b> | <b>Pr &gt;  t </b> |
|------------------|----------------|-----------------|-----------------------|-----------|----------------|--------------------|
| <b>Intercept</b> | <b>2</b>       | 0.3561          | 0.3484                | 478       | 1.02           | 0.3073             |
| <b>Intercept</b> | <b>3</b>       | 0.8540          | 0.3439                | 478       | 2.48           | 0.0134             |
| <b>Intercept</b> | <b>4</b>       | 1.0727          | 0.4371                | 478       | 2.45           | 0.0145             |
| <b>treat</b>     | <b>2</b>       | 0.1551          | 0.5047                | 0         | 0.31           | .                  |
| <b>treat</b>     | <b>3</b>       | 0.3272          | 0.4900                | 0         | 0.67           | .                  |
| <b>treat</b>     | <b>4</b>       | -0.04118        | 0.6205                | 0         | -0.07          | .                  |
| <b>occasion</b>  | <b>2</b>       | -0.4227         | 0.4338                | 0         | -0.97          | .                  |
| <b>occasion</b>  | <b>3</b>       | -0.8486         | 0.4211                | 0         | -2.02          | .                  |

### Solutions for Fixed Effects

| Effect         | outcome | Estimate | Standard Error | DF | t Value | Pr >  t |
|----------------|---------|----------|----------------|----|---------|---------|
| occasion       | 4       | -2.4529  | 0.5660         | 0  | -4.33   | .       |
| treat*occasion | 2       | 0.1143   | 0.6057         | 0  | 0.19    | .       |
| treat*occasion | 3       | -1.2945  | 0.6210         | 0  | -2.08   | .       |
| treat*occasion | 4       | -1.6081  | 0.7836         | 0  | -2.05   | .       |

### Type III Tests of Fixed Effects

| Effect         | Num DF | Den DF | F Value | Pr > F |
|----------------|--------|--------|---------|--------|
| treat          | 3      | 0      | 0.23    | .      |
| occasion       | 3      | 0      | 6.62    | .      |
| treat*occasion | 3      | 0      | 3.31    | .      |